{% extends "base.html" %} {% block extra_head %} {% endblock %} {% block content %}

Documentation

Cost Estimation API

Endpoint

```
POST /api/estimate
```

Request Format

```
{
  "project_state": "GA",
  "project_type": "Commercial",
  "construction_category": "Office",
  "cnt_division": 10,
  "cnt_item_code": 100,
  "cnt_csi_grp_unq": 20,
  "acf": 1.0
}
```

Field	Type	Description	Required
project_state	String	US State code (e.g., "GA", "CA")	Yes
project_type	String	Type of construction project (e.g., "Commercial")	Yes
construction_category	String	Specific construction category (e.g., "Office")	Yes
cnt_division	Integer	Number of project divisions	No
cnt_item_code	Integer	Number of item codes	No
cnt_csi_grp_unq	Integer	Number of unique CSI groups	No
acf	Float	Area Cost Factor (0.7-1.5)	No

Response Format

```
{
  "estimated_cost": 5207578.25,
  "confidence_interval": {
    "low": 4063473.58,
    "high": 6351683.00
  },
  "model_metrics": {
    "mape": 21.97,
    "rmse": 271543.90,
    "r2": 0.9463,
    "model_type": "Random Forest"
  }
}
```

Field	Type	Description
estimated_cost	Float	The estimated project cost in USD
confidence_interval.low	Float	Lower bound of the confidence interval
confidence_interval.high	Float	Upper bound of the confidence interval
model_metrics.mape	Float	Mean Absolute Percentage Error of the model
model_metrics.rmse	Float	Root Mean Square Error of the model
model_metrics.r2	Float	R-squared value (coefficient of determination)
model_metrics.model_type	String	The machine learning model used for prediction

Example API Call

```
curl -X POST https://example.com/api/estimate \
-H "Content-Type: application/json" \
-d '{
  "project_state": "GA",
  "project_type": "Commercial",
  "construction_category": "Office",
  "cnt_division": 10,
  "cnt_item_code": 100,
  "cnt_csi_grp_unq": 20,
```

```
  "acf": 1.0
}'
```


User Guide

Using the Cost Estimator

1. Navigate to the "Cost Estimator" page
2. Fill in the project details:
   - Select the project state
   - Choose the project type
   - Select the construction category
   - Enter the number of divisions
   - Enter the number of item codes
   - Enter the number of CSI groups
   - Enter the area cost factor (ACF)
3. Click "Estimate Cost" to get the cost prediction

Interpreting Results

- **Estimated Cost:** The model's best prediction for the total project cost
- **Confidence Interval:** A range in which the actual cost is likely to fall
- **MAPE:** Mean Absolute Percentage Error - lower is better
- **Model Type:** The type of model used for this prediction

Note: This estimator provides Class 5 estimates suitable for conceptual planning and feasibility studies.

Final Project Report

View our comprehensive project report with detailed methodology, data analysis, and model evaluation.

{% if 'final_report.pdf' in report_files %} [View Final Report] {% else %} [View Final Report] {% endif %}

Model Information

Random Forest Model

Our cost estimation model uses a Random Forest algorithm, which was selected after extensive testing and comparison with other models. Random Forests excel at handling non-linear relationships and categorical variables, making them ideal for construction cost estimation.

Key Performance Metrics:

{{ metrics.mape }}%
MAPE
${{ "{:,.2f}".format(metrics.rmse) }}
RMSE
{{ metrics.r2 }}
R^2

Data Sources

The model was trained on historical construction project data from 2010-2025, normalized to 2025 dollars. The dataset includes over 6,000 commercial, residential, and institutional projects across all 50 states.

Feature Importance:

- Project State (Geographic Location)
- Construction Category
- Project Type
- Number of Divisions
- Number of Item Codes

Project Team

{% for member in team %}
{% if member.name in ['Krishna Aryal', 'Kumar Sawan'] %} {% elif member.name == 'Neema Kafwimi' %} {% else %} {% endif %}

{{ member.name }}

{{ member.role }}

{{ member.get('description', 'Georgia Institute of Technology') }}
{% endfor %}
{% endblock %}